EXP NO-7
DATE-

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM: To write a program to implement flow control at the Data link layer using the Sliding Window protocol.

ALGORITHM:

1) Start the program.
2) Read the total no. of frames (N) window Size(W), the frame number to be lost.
3) Initialize:
   base=0 - first window ledged frame.
   next seq=0 - Next frame to be sent.
4) While base <N, determine the sender window from base to base + W-1
5) Send all frames within the current window and increment next seq after sending each frame
6) If the specified frame is lost:
   * Receiver deletes the missing frame.
   * Receiver discards the out-of-order frame.
   * Sender experience a time out.
7) If there is no frame loss:
   * Receiver accepts the frames in order.
   * Receiver sends on for each received frame.
8) Repeat step 4-7 until all frames are successfully transmitted acknowledged.
9) Display that all frames
10) Stop the program.

CODE:

```
import java.util.scanner;
Public class Sliding window{
Static void simulate (int total frames,
int window size, Integer lost Frame){
 int base=0; int nextseq=0;
while (Base < totalframes){
    int window End = Math.min( , base +
                window size -1);
System.out.println ("Sender Window :
 (frame" + Base " -frame" + WindowEnd+"));
while (nextseq < base + windowsize &&
     nextseq - totalFrames){
System.out.println("Node A -> Sending Frame"
                +next seq);
if(last Frame != null && nextseq == = Last frame)
{ System.out.println ("Frame" + next seq +
          "Loss in transit!") ; } next seq; }
if (lost Frame! = null && Base <= lost Frame
 && last Frame < next seq){
system.out.println("Node B -> Frame" +
     lost Frame + "missing ");
System.out.println("Node A -> Time out!
  Resending window starting from Frame" +
            lost Frame" : " );
  next seq = lost frame;
  lost frame = = null;
   continue;
for ( int f = base; f < next seq; f++){
System.out.println ("Node B -> Received Frame" +
  f + , Sending Ack "+f); }
Base = next seq ; ); } }
Public static void main(String[ ] args){
 Scanner Sc = new Scanner(
```

```
System.out.println ("Enter total number of frames
        to send");
int total frames = sc.nextInt();
System.out.println ("Enter frame numbers to
    simulate as Lost \n" + "(0 to "+(total frames-1)
      + " , enter -1 for no loss ): " ); }else{
  lost Frame = input lost Frame ; }
Simulate (total frame, Window Size, lost Frame);
    Sc.close (); } }
```

OUTPUT:

```
Enter total number of frames to send : 3
Enter Senders window Size : 2
Enter frame number to simulate as lost
  (0 to 2 , enter -1 for no loss): 1

Sender Window: [Frame 0 - Frame 1]
    Node A -> Sending Frame 0
    Node B -> Sending Frame 1
        (Frame 1 Lost in transit)

    Node B -> Frame 1 missing
    Node A -> Timeout ! Resending frame frame 1'
Sender Window : [Frame 1 - Frame 2]
    Node A -> Sending Frame 1
    Node A -> Sending Frame 2
    Node B -> Received Frame 1, Sending Ack
All 3 frames successfully transmitted from
Node A to Node B.
```

RESULT: Thus, the program to implement flow control at the Data link layer using the Sliding Window protocol was implemented successfully!